5.5 hours on examples

Problem 2 (Snake3)

We have generating function,

$$F(X) = \sum_{n\geq 0}\left(\sum_{k\geq 0}\binom{n}{k}\binom{k}{m}\right)x^n$$

$$= \sum_{k\geq 0}\binom{k}{m}\sum_{n\geq 0}\binom{n}{k}x^n$$

$$= \sum_{k\geq 0}\binom{k}{m}\frac{x^k}{(1-x)^{k+1}}$$

$$= \frac{1}{1-x}\cdot\frac{\left(\frac{x}{1-x}\right)^m}{\left(1-\frac{x}{1-x}\right)^{m+1}}$$

$$= \frac{(x)^m}{((1-x)-x)^{m+1}}$$

$$= \frac{1}{2^m}\cdot\frac{(2x)^m}{(1-2x)^{m+1}}$$

$$= \frac{1}{2^m}\sum_{n\geq 0}\binom{n}{m}(2x)^n$$

Therefore, we have $2^{n-m}\binom{n}{m}$.

Problem 3 (Snake1)

We have generating function,

$$F(X) = \sum_{n\geq 0}\left(\sum_{k\geq 0}\binom{n+k}{m+2k}\binom{2k}{k}\frac{(-1)^k}{k+1}\right)X^n$$

Note that $n+k\geq m+2k \Leftrightarrow n\geq m+k$ for $\binom{n+k}{m+2k}\neq 0$.

$$= \sum_{k\geq 0}\frac{(-1)^k}{k+1}\binom{2k}{k}\sum_{n\geq m+k}\binom{n+k}{m+2k}X^n$$

We let $n' = n-m-k$.

$$= \sum_{k\geq 0}\frac{(-1)^k}{k+1}\binom{2k}{k}\sum_{n'\geq 0}\binom{n'+m+k+k}{m+2k}X^{n'+m+k}$$

$$= \sum_{k\geq 0} \frac{(-1)^k}{k+1} \binom{2k}{k} X^{\mathrm{m}+k} \sum_{n'\geq 0} \binom{n' + m + 2k}{m + 2k} X^{n'}$$

$$= \sum_{k\geq 0} \frac{(-1)^k}{k+1} \binom{2k}{k} \frac{X^{m+k}}{(1-X)^{m+2k+1}}$$

$$= \frac{X^m}{(1-X)^{m+1}} \sum_{k\geq 0} \frac{1}{k+1} \binom{2k}{k} \left(\frac{-X}{(1-X)^2}\right)^k$$

$$= \frac{X^m}{(1-X)^{m+1}} \cdot \frac{1 - \sqrt{1 + \frac{4X}{(1-X)^2}}}{\frac{-2X}{(1-X)^2}}$$

$$= \frac{X^m}{(1-X)^{m+1}} \cdot \frac{\frac{X+1}{1-X} - 1}{\frac{2X}{(1-X)^2}}$$

$$= \frac{X^m}{(1-X)^{m+1}} \cdot (1-X)$$

$$= X \sum_{n\geq 0} \binom{n}{m-1} x^n$$

$$= \sum_{n\geq 1} \binom{n-1}{m-1} x^n$$

The answer is hence $\binom{n-1}{m-1}$.

7 hours, 4 clubs

Problem 5 (23CMIMCC7)

We have the generating function,

$$F(X) = \prod_{i=1}^{100} \left(\frac{2(i+1)^2 - 1}{2(i+1)^2} + \frac{x}{2(i+1)^2} \right)$$

We want the sum of the coefficient of terms with odd degrees. We can use the roots of unity filter to get this is,

$$N = \frac{F(1) - F(-1)}{2}$$

Obviously, $F(1) = 1$.

$$F(-1) = \prod_{i=1}^{100} \left(\frac{2(i+1)^2 - 1}{2(i+1)^2} + \frac{-1}{2(i+1)^2} \right) = \prod_{i=1}^{100} \left(\frac{(i)(i+2)}{(i+1)^2} \right) = \frac{1 \cdot 2 \cdot 101 \cdot 102}{2^2 \cdot 101^2} = \frac{51}{101}$$

$$N = \frac{25}{101}$$

Problem 6 (HMMT 2007 C9)

Our generating function is,

$$F(X) = (X + 2X^2 + 3X^3 + 4X^4 + \cdots)(X + 2X^2 + 3X^3 + 4X^4 + \cdots)(X + 2X^2 + 3X^3 + 4X^4 + \cdots)$$

$$= (X + 2X^2 + 3X^3 + 4X^4 + \cdots)^3$$

$$= \left(\frac{d}{dx}(1 + X + \cdots) \cdot X\right)^3 = \frac{X^3}{(1-X)^6}$$

$$= X^3 \sum_{k\geq 0} \binom{-6}{k} (-1)^k X^k = X^3 \sum_{k\geq 0} \binom{5+k}{k} X^k$$

Therefore, the coefficient of the x^n is $\binom{n+2}{n-3}$. The answer is $\binom{19}{14}$.

Problem 7 (Z0A5DBDB)

Our generating function is,

$$F(X) = \sum_{n\geq 0} \left(\sum_{a+b=m} \binom{2a}{a}\binom{2b}{b} \right) X^n$$

$$= \sum_{a\geq 0} \binom{2a}{a} \sum_{n\geq a} \binom{2n-2a}{n-a} X^n$$

$$= \sum_{a\geq 0} \binom{2a}{a} (1-4x)^{-\frac{1}{2}} X^a$$

$$= \frac{1}{1-4x}$$

We're hence done.

Problem 9 (HMMT 2013)

Note that generating function,

$$F(X) = \left(\frac{1}{1-\frac{1}{3}X}\right)^7 = \left(1 + \frac{1}{3}X + \frac{1}{3^2}X^2 + \cdots\right)^7$$

$$= \sum_{a_1=0}^{\infty} \sum_{a_2=0}^{\infty} \cdots \sum_{a_7=0}^{\infty} \frac{X^{a_1+a_2+\cdots+a_7}}{3^{a_1+a_2+\cdots+a_7}}$$

We differentiate both sides,

$$F'(X)=\sum_{a_1=0}^{\infty}\sum_{a_2=0}^{\infty}\dots\sum_{a_7=0}^{\infty}\frac{(a_1+a_2+\cdots+a_7)\cdot X^{a_1+a_2+\cdots+a_7-1}}{3^{a_1+a_2+\cdots+a_7}}$$
$$=\frac{d}{dX}\left(\left(\frac{3}{3-X}\right)^7\right)=\frac{d}{dX}(3^7\cdot(3-X)^{-7})$$
$$=3^7\cdot 7(3-X)^{-8}$$

We plug in $X=1$ to get,

$$F'(X)=\frac{3^7\cdot 7}{2^8}=\sum_{a_1=0}^{\infty}\sum_{a_2=0}^{\infty}\dots\sum_{a_7=0}^{\infty}\frac{a_1+a_2+\cdots+a_7}{3^{a_1+a_2+\cdots+a_7}}$$

9.5 hours, 16 clubs

Problem 10 (HMMT 2020 C10)

$$F(x)=0.04+0.04\left(0.48x+\frac{0.48}{x}\right)+0.04\left(0.48x+\frac{0.48}{x}\right)^2+\cdots$$
$$=\frac{0.04}{1-\left(0.48x+\frac{0.48}{x}\right)}=\frac{1}{25-12x-\frac{12}{x}}=\frac{-x}{12x^2-25x+12}=\frac{\frac{1}{7}}{1-\frac{3x}{4}}+\frac{-\frac{1}{7}}{1-\frac{4x}{3}}$$
$$=\frac{\frac{1}{7}}{1-\frac{3x}{4}}+\frac{-\frac{1}{7x}}{\frac{1}{x}-\frac{4}{3}}=\frac{\frac{1}{7}}{1-\frac{3x}{4}}+\frac{\frac{3}{28x}}{1-\frac{3}{4x}}$$
$$=\cdots\frac{9}{112}x^{-2}+\frac{3}{28}x^{-1}+\frac{1}{7}+\frac{3}{28}x+\frac{9}{112}x^2+\cdots$$

Our sum is hence,

$$E=2\cdot\left(\frac{1}{7}\cdot\left(\frac{3}{4}\right)^1\cdot 1+\frac{1}{7}\cdot\left(\frac{3}{4}\right)^2\cdot 2+\frac{1}{7}\cdot\left(\frac{3}{4}\right)^3\cdot 3+\cdots\right)$$
$$=\frac{2}{7}\cdot\left(\frac{\frac{3}{4}}{1-\frac{3}{4}}+\frac{\frac{3^2}{4}}{1-\frac{3}{4}}+\frac{\frac{3^3}{4}}{1-\frac{3}{4}}+\cdots\right)$$
$$=\frac{32}{7}\cdot\frac{3}{4}=\frac{24}{7}$$

Problem 13 (AIME 2018)

The generating function is,

$$F(X)=\left(\frac{1}{2}+\frac{1}{2}X\right)\left(\frac{1}{2}+\frac{1}{2}X^2\right)\dots\left(\frac{1}{2}+\frac{1}{2}X^{18}\right)=\frac{1}{2^{18}}(1+X)(1+X^2)\dots(1+X^{18})$$

We want the sum of coefficients of the degree multiple 3 terms. We use roots of unity filter,

$$= \frac{F(1) + F\left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right) + F\left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i\right)}{2^{18} \cdot 3}$$

$$= \frac{2^{18} + 2^6\left(1 + \left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)\right)^6\left(1 + \left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i\right)\right)^6 + 2^6\left(1 + \left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)\right)^6\left(1 + \left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i\right)\right)^6}{2^{18} \cdot 3}$$

$$= \frac{2^{18} + 2^7}{2^{18} \cdot 3} = \frac{\frac{2048 + 1}{3}}{2048} = \frac{683}{2048}$$

11.5 hours, 23 clubs

<u>Problem 14 (Putnam 1990)</u>

We casework on $i = |A|$ and $j = |B|$. Our sum is,

$$\sum_{i+j \leq n} \binom{n-j}{i}\binom{n-i}{j}$$

Our generating function is,

$$F(x) = \sum_{n \geq 0}\left(\sum_{i+j \leq n}\binom{n-j}{i}\binom{n-i}{j}\right)x^n = \sum_{i \geq 0}\sum_{j \geq 0}\sum_{n \geq i+j}\binom{n-j}{i}\binom{n-i}{j}x^n$$

$$= \sum_{i \geq 0}\sum_{j \geq 0} x^{i+j}\sum_{n \geq 0}\binom{n+i}{i}\binom{n+j}{j}x^n$$

$$= \sum_{i \geq 0}\sum_{j \geq 0}(-x)^{i+j}\sum_{n \geq 0}\binom{-n-1}{i}\binom{-n-1}{j}x^n$$

$$= \sum_{n \geq 0} x^n \sum_{i \geq 0}(-x)^i\binom{-n-1}{i}\sum_{j \geq 0}(-x)^j\binom{-n-1}{j}$$

$$= \sum_{n \geq 0} x^n\,(1-x)^{-2n-2} = \frac{\sum_{n \geq 0}\left(\frac{x}{(1-x)^2}\right)^n}{(1-x)^2}$$

$$= \frac{\frac{1}{1 - \frac{x}{(1-x)^2}}}{(1-x)^2} = \frac{1}{x^2 - 3x + 1} = \frac{\frac{3+\sqrt{5}}{2\sqrt{5}}}{1 - \frac{3+\sqrt{5}}{2}x} - \frac{\frac{3-\sqrt{5}}{2\sqrt{5}}}{1 - \frac{3-\sqrt{5}}{2}x}$$

The answer is hence $\frac{1}{\sqrt{5}}\left(\left(\frac{3+\sqrt{5}}{2}\right)^{n+1} - \left(\frac{3-\sqrt{5}}{2}\right)^{n+1}\right)$.

Problem 17 (HMMT 2017 C8)

Our generating function is as follows. The degree term of x_i is the degree of vertex i.

$$f(x_0,\ x_1, x_2, \dots,\ x_{15}) = \prod_{0\le i<j<16} \left(\frac{1}{2}+\frac{1}{2}x_i x_j\right)$$

We use the roots of unity filter.

$$N = \frac{1}{4^{16}} \sum_{a_0=0}^{3} \sum_{a_1=0}^{3} \dots \sum_{a_{15}=0}^{3} f(\zeta^{a_0},\ \zeta^{a_1},\ \dots,\ \zeta^{a_{15}})$$
$$= \frac{1}{4^{16}} \sum_{a_0=0}^{3} \sum_{a_1=0}^{3} \dots \sum_{a_{15}=0}^{3} f(\zeta^{a_0},\ \zeta^{a_1},\ \dots,\ \zeta^{a_{15}})$$

Note that if we have any pair $(a_i,\ a_j)$ for $i \ne j$ where $a_i + a_j = 2$, there'd be a 0 in the product and it'd have no contribution to the sum. Therefore, we only count the combinations $a_0 \dots a_{15}$ which have no $(1,\ 1)$, $(3,\ 3)$, or $(0,\ 2)$.

Case 1: 1 case of $a_i = 0$,

$$\prod_{0\le i<j<16} \left(\frac{1}{2}+\frac{1}{2}\zeta^0\zeta^0\right) = 1$$

Case 2: 1 case of $a_i = 2$,

$$\prod_{0\le i<j<16} \left(\frac{1}{2}+\frac{1}{2}\zeta^{a_i}\zeta^{a_j}\right) = \prod_{0\le i<j<16} \left(\frac{1}{2}+\frac{1}{2}\zeta^2\zeta^2\right) = \prod 1 = 1$$

Case 3: 16 permutations of $a_0 = 1; a_1 = a_2 = \dots = a_{16} = 0$

$$16 \prod_{0\le i<j<16} \left(\frac{1}{2}+\frac{1}{2}\zeta^{a_i}\zeta^{a_j}\right) = 16\left(\frac{1}{2}+\frac{i}{2}\right)^{15} = \frac{1}{16}-\frac{1}{16}i$$

Case 4: 16 permutations of $a_0 = 1; a_1 = a_2 = \dots = a_{16} = 2$

$$16 \prod_{0\le i<j<16} \left(\frac{1}{2}+\frac{1}{2}\zeta^{a_i}\zeta^{a_j}\right) = 16\left(\frac{1}{2}-\frac{i}{2}\right)^{15} = \frac{1}{16}+\frac{1}{16}i$$

Case 5: 16 permutations of $a_0 = 3; a_1 = a_2 = \dots = a_{16} = 0$

$$16 \prod_{0\le i<j<16} \left(\frac{1}{2}+\frac{1}{2}\zeta^{a_i}\zeta^{a_j}\right) = 16\left(\frac{1}{2}-\frac{i}{2}\right)^{15} = \frac{1}{16}+\frac{1}{16}i$$

Case 6: 16 permutations of $a_0 = 3; a_1 = a_2 = \dots = a_{16} = 2$

$$16 \prod_{0\le i<j<16} \left(\frac{1}{2}+\frac{1}{2}\zeta^{a_i}\zeta^{a_j}\right) = 16\left(\frac{1}{2}+\frac{i}{2}\right)^{15} = \frac{1}{16}-\frac{1}{16}i$$

Case 7: $16 \cdot 15$ permutations of $a_0 = 1; a_1 = 3; a_2 = \dots = a_{15} = 0$

$$15 \cdot 2^4 \prod_{0\le i<j<16} \left(\frac{1}{2}+\frac{1}{2}\zeta^{a_i}\zeta^{a_j}\right) = 15 \cdot 2^4 \left(\frac{1}{2}+\frac{i}{2}\right)^{14}\left(\frac{1}{2}-\frac{i}{2}\right)^{14} = \frac{15}{1024}$$

Case 8: $16 \cdot 15$ permutations of $a_0 = 1; a_1 = 3; a_2 = \cdots = a_{15} = 2$

$$15 \cdot 2^4 \prod_{0 \le i < j < 16} \left(\frac{1}{2} + \frac{1}{2}\zeta^{a_i}\zeta^{a_j}\right) = 15 \cdot 2^4 \left(\frac{1}{2} - \frac{i}{2}\right)^{14} \left(\frac{1}{2} + \frac{i}{2}\right)^{14} = \frac{15}{1024}$$

The sum is $1 + 1 + \frac{1}{4} + \frac{15}{512} = \frac{1167}{512}$.

$$N = \frac{1}{4^{16}} \cdot \frac{1167}{512} = \frac{1167}{2^{41}}$$

13.5 hours; 37 clubs